
EXERCISE

Lecture 2: Probability Recap

AI508 — Algorithmic Game Theory

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1 Probability spaces, outcomes, and events

A probability model consists of a sample space Ω of possible outcomes ω together with probabilities assigned to events A .

Definition: Outcome and event

An *outcome* is a single element $\omega \in \Omega$. And *event* is a subset $A \subseteq \Omega$

Example: Different sample spaces

If two fair coins are tossed, then

$$\Omega = \{HH, HT, TH, TT\}$$

An event “exactly one head” would then be

$$HT, TH$$

For a finite or countable sample space, probabilities satisfy

$$\mathbb{P}(A) = \sum_{\omega \in A} \mathbb{P}(\omega), \quad \sum_{\omega \in \Omega} = 1.$$

Remember that

$$P(A^C) = 1 - P(A)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

2 Random variables

Definition: Random Variable

The random variable X is a function that maps

$$X : \Omega \rightarrow \mathbb{R}.$$

Example: Coin

Two coins giving heads

$$X = \{0, 1, 2\}$$

Where,

- $0 = X\{TT\},$
- $1 = X\{TH\} = X\{HT\},$
- $2 = X\{HH\}$

3 Discrete vs continuous random variables

The distinction between discrete and continuous refers to the *distribution of X* , not to whether the underlying sample space Ω is discrete or continuous.

$$\mathbb{E}[X] = \sum_{x \in X} x \cdot P(X = x)$$

Example: Continuous sample space, discrete random variable

Given the uniform probability distribution $\Omega = [0, 1]$, then

$$X(\omega) = \begin{cases} -1 & w \leq \frac{1}{2} \\ 1 & w > \frac{1}{2} \end{cases}$$

Although Ω is continuous, X takes only two values. Hence X is *discrete*, with

$$\mathbb{P}(X = -1) = \mathbb{P}(X = 1) = 1/2$$

Example: Continuous random variable on the same space

Again let $\Omega = [0, 1]$ be uniform, but now define

$$X(\omega) = \Omega$$

Then $X \sim \text{Uniform}[0, 1]$, so X is a continuous random variable.

For a discrete random variable, we write its probability mass function as

$$p_X(x) = \mathbb{P}(X = x)$$

Where the probability satisfies

$$\sum_x p_X(x) = 1$$

4 Joint distributions and marginalization

Definition: Joined distribution

Given that X, Y are random variables defined on the same probability space. Then their joint distribution is

$$p_{XY}(x, y) = \mathbb{P}(X = x, Y = y)$$

Definition: Marginalization

If we know the joint distribution but only care about X , we sum out the possible values of Y :

$$p_X(x) = \mathbb{P}(X = x) = \sum_{y \in Y} p_{XY}(x, y)$$

And similarly

$$p_Y(y) = \mathbb{P}(Y = y) = \sum_{x \in X} p_{XY}(x, y)$$

Example: Marginalization

Suppose that,

	$Y = 0$	$Y = 1$
$X = 0$	0.2	0.3
$X = 1$	0.1	0.4

Table 1: Example table

Then

$$\mathbb{P}(X = 0) = 0.2 + 0.3 = 0.5, \quad \mathbb{P}(X = 1) = 0.1 + 0.4 = 0.5$$

5 Expectation

Definition: Expectation of a random variable

For a discrete random variable X , its expectation is

$$\mathbb{E}[X] = \sum_x x\mathbb{P}(X = x)$$

whenever the sum is well defined. For a countably supported random variable, a sufficient condition is $\mathbb{E}[X] < \infty$.

Example: Expectation of a discrete random variable

If X takes values $-1, 0, 1$ uniformly, then

$$\mathbb{E}[X^2] = (-1)^2 \frac{1}{3} + 0^2 \frac{1}{3} + 1^2 \frac{1}{3} = \frac{2}{3}$$

We did not need to separately compute the distribution of X^2

Definition: The lazy statistician rule (discrete)

Suppose we want the expectation of a function $f(X)$. We do not need to first derive the distribution of $f(X)$. Instead,

$$\mathbb{E}[f(X)] = \sum_x f(x)\mathbb{P}(X = x)$$

for a discrete random variable. This identity is commonly called the *Law of the Unconscious Statistician* (LOTUS); informally, it is the “lazy statistician rule.”

Proof: LOTUS

To see why it holds, start from the definition of the expectation of $f(X)$:

$$\mathbb{E}[f(X)] = \sum_z z\mathbb{P}(f(X) = z)$$

Now group together all values x for which $f(x) = z$:

$$\mathbb{P}(f(X) = z) = \sum_x (\mathbb{P}(X = x)) \quad \text{where } x : f(x) = z$$

So...

$$\begin{aligned} \mathbb{E}[f(X)] &= \sum_z z \sum_x (\mathbb{P}(X = x)) \quad \text{where } x : f(x) = z \\ &= \sum_x f(x)\mathbb{P}(X = x). \end{aligned}$$

□

Definition: The lazy statistician rule (continuous)

Suppose we want the expectation of a function $f(X)$. We do not need to first derive the distribution of $f(X)$. Instead,

$$\mathbb{E}[f(X)] = \int_{-\infty}^{\infty} f(x) f_X(x) dx$$

for a continuous random variable with density f_X .

Definition: Bernoulli

Given that $X = \begin{cases} 1 & p \\ 0 & 1-p \end{cases}$

So that $\mathbb{E}[X] = 1p + 0(1-p) = p$

Theorem: Indicators and counting

An indicator random variable for an event A is

$$\mathbb{1}_A = \begin{cases} 1, & A \text{ occurs,} \\ 0, & A \text{ does not occur.} \end{cases}$$

Since $\mathbb{1}_A$ is Bernoulli,

$$\mathbb{E}[\mathbb{1}_A] = \mathbb{P}(A)$$

This makes indicator variables extremely useful for counting.

6 Variance and covariance

Definition: Variance

Variance captures the spread of value around the

$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - \mathbb{E}[X]^2$$

Example: Variance of a Bernoulli

Given X where

$$X = \begin{cases} 1 & p \\ 0 & 1-p \end{cases} \quad \text{with } \mathbb{E}[X] = p$$

,

$$\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = (1^2p + 0^2(1-p)) - p^2 = p - p^2 = p(1-p)$$

Definition: Covariance

Given X, Y ,

$$\begin{aligned}\text{Cov}(X, Y) &= \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] \\ &= \mathbb{E}[XY + \mathbb{E}[X]\mathbb{E}[Y] - \mathbb{E}[Y] - Y\mathbb{E}[X]] \\ &= \mathbb{E}[XY] + \mathbb{E}[X]\mathbb{E}[Y] - \mathbb{E}[X]\mathbb{E}[Y] - \mathbb{E}[Y\mathbb{E}[X]] \\ &= \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]\end{aligned}$$

Definition: Uncorrelation of 2 random variables

2 R.V's are uncorrelated if

$$\text{Cov}(X, Y) = 0 \implies \mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$$

This does not generally mean $\mathbb{E}[XY] = 0$ unless at least one of the means is zero.

6.1 Independence implies zero covariance

Definition: Independence

X, Y are independent if

$$p_{XY}(x, y) = p_X(x)p_Y(y) \quad \forall (x, y)$$

If X and Y are independent, then their joint mass function factorizes:

$$p_{XY}(x, y) = p_X(x)p_Y(y)$$

Therefore

Proof

$$\begin{aligned}\mathbb{E}[XY] &= \sum_x \sum_y xyp_{XY}(x, y) \\ &= \sum_x \sum_y xyp_X(x)p_Y(y) \\ &= \left(\sum_x xp_X(x) \right) \left(\sum_y yp_Y(y) \right) \\ &= \mathbb{E}[X]\mathbb{E}[Y]\end{aligned}$$

Because of this

$$X \perp Y \implies \text{Cov}(X, Y) = 0$$

□

The converse is false in general.

6.2 Gaussian exception

For jointly Gaussian random variables, zero covariance does imply independence. Thus if (X, Y) is **jointly Gaussian**,

$$\text{Cov}(X, Y) = 0 \iff X \perp Y$$

The assumption that they are *jointly* Gaussian is important; it is not enough for the two marginal distributions to be Gaussian separately.

7 Conditional probability

Definition: Conditional probability

Assume there are 2 events A and B and let $P(B) > 0$

Then the probability of event A given B is:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

Conditioning means that we restrict our attention to outcomes in which B has occurred and renormalize probabilities within that restricted set.

Example: Conditional probability

Given $B = \{x > 3\} = \{4, 5, 6\}$

The probability that $A = \text{EVEN} = \{2, 4, 6\}$ is:

$$\frac{P(A \cap B)}{P(B)} = \frac{\frac{2}{6}}{\frac{3}{6}} = \frac{2}{3}$$

Example: Conditional probability with random variables

$$\mathbb{P}(X = x \mid Y = y) = \frac{\mathbb{P}(X = x, Y = y)}{\mathbb{P}(Y = y)}, \quad \mathbb{P}(Y = y) > 0.$$

Definition: Law of total probability

If $B_1, \dots, B_m$ partition the sample space and $\mathbb{P}(B_i) > 0$, then

$$\mathbb{P}(A) = \sum_{i=1}^m \mathbb{P}(A \mid B_i) \mathbb{P}(B_i)$$

Indeed,

$$\begin{aligned} \mathbb{P}(A) &= \sum_{i=1}^m \mathbb{P}(A \cap B_i) \\ &= \sum_{i=1}^m \mathbb{P}(A \mid B_i) \mathbb{P}(B_i) \end{aligned}$$

Definition: Bayes Rule

Given A, B

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(B \mid A)\mathbb{P}(A)}{\mathbb{P}(B)}$$

Where $\mathbb{P}(B \mid A)$ is the *likelihood*, $\mathbb{P}(A)$ is the *prior* and $\mathbb{P}(B)$ is the *marginal evidence*

A useful interpretation

$$\text{posterior} = \frac{\text{likelihood} \times \text{prior}}{\text{evidence}}$$

Definition: Independent events of a random variable

Event A and B are independent if and only if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$$

Theorem: A and A^c

If A and A^c are the two possibilities, then

$$\mathbb{P}(B) = \mathbb{P}(B \mid A)\mathbb{P}(A) + \mathbb{P}(B \mid A^c)\mathbb{P}(A^c)$$

Thus the denominator normalizes the updated belief by accounting for all ways in which the evidence could have arisen.

8 Conditional expectation

Note that conditional expectations are not scalars,

Definition: Conditional expectation

Given Y

$$\mathbb{E}[X \mid Y = y] = \sum_{x \in X} x \cdot P(X = x \mid Y = y)$$

Theorem: Law of Total Expectation

Given X and Y ,

$$\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X \mid Y = y]] = \sum_{y \in Y} \left(\sum_{x \in X} x \cdot P(X = x \mid Y = y) \right) P(Y = y)$$

Proof of LoTE

Starting from the definition,

$$\mathbb{E}[X] = \sum_x x \mathbb{P}(X = x)$$

Now marginalize over Y

$$\mathbb{P}(X = x) = \sum_y \mathbb{P}(X = x, Y = y)$$

Therefore

$$\begin{aligned} \mathbb{E}[X] &= \sum_x x \sum_y \mathbb{P}(X = x, Y = y) \\ &= \sum_y \sum_x x \mathbb{P}(X = x, Y = y) \\ &= \sum_y \sum_x x \mathbb{P}(X = x \mid Y = y) \mathbb{P}(Y = y) \\ &= \sum_y P(Y = y) \underbrace{\sum_x x \mathbb{P}(X = x \mid Y = y)}_{\mathbb{E}[X \mid Y=y]} \\ &= \sum_y \mathbb{E}[X \mid Y = y] \mathbb{P}(Y = y) \\ &= \mathbb{E}[\mathbb{E}[X \mid Y]] \end{aligned}$$

For countable supports, the same proof works whenever the relevant sums are absolutely convergent, for example when $\mathbb{E}[X] < \infty$.

9 Continuous random variables

A continuous random variable is described by a probability density function f_X satisfying

$$f_X(x) \geq 0, \quad \int_{-\infty}^{\infty} f_X(x) \, dx = 1.$$

Probabilities are then obtained by integration:

$$\mathbb{P}(a \leq X \leq b) = \int_a^b f_X(x) \, dx$$

For a continuous random variable X ,

$$\mathbb{P}(X = x) = 0$$

for every individual point x .

Definition: Expectation of a continuous random variable

The expectation of a continuous random variable is

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x) \, dx$$

Definition: LOTUS, the continuous version

The continuous version of LOTUS is

$$\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) \, dx$$

For jointly continuous X, Y with joint density $f_{X,Y}$, marginalization becomes

$$f_X(x) \int_{-\infty}^{\infty} f_{X,Y}(x, y) \, dy$$

10 Gaussian distributions

10.1 Univariate Gaussian

A Gaussian random variable with mean μ and variance σ^2 is written as:

$$X \sim \mathcal{N}(\mu, \sigma^2)$$

and has density

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

The mean μ controls the center, while σ^2 controls the spread.

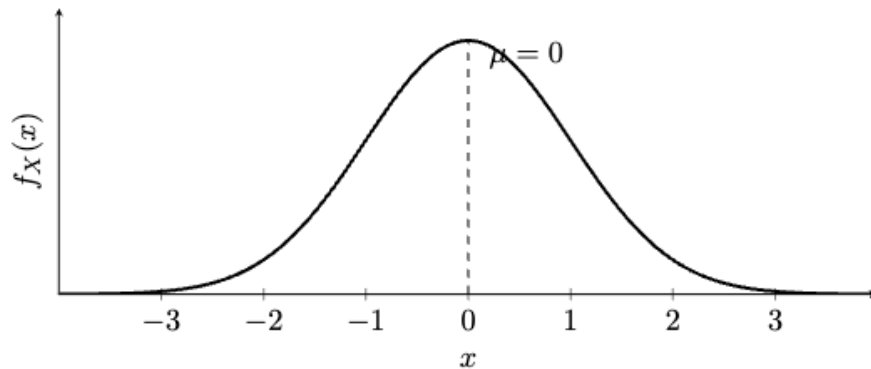


Figure 1: The density of a standard Gaussian $\mathcal{N}(0, 1)$.

10.2 Multivariate Gaussian

A random vector $X \in \mathbb{R}^d$ is multivariate Gaussian with mean vector $\mu \in \mathbb{R}^d$ and positive-definite *covariance*, matrix $\Sigma \in \mathbb{R}^{d \times d}$ if

$$X \sim \mathcal{N}(\mu, \Sigma)$$

with density

$$f_X(x) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(x-\mu)^\top \Sigma^{-1}(x-\mu)\right).$$

The covariance matrix controls both the spread in each coordinate and the dependence between coordinates.

In 2D dimensions

$$\Sigma = \begin{bmatrix} \sigma_1^2 & \text{Cov}(X_1, X_2) \\ \text{Cov}(X_1, X_2) & \sigma_2^2 \end{bmatrix}$$

If the off-diagonal is zero, the contours are axis-aligned. A nonzero covariance rotates the elliptical contours.

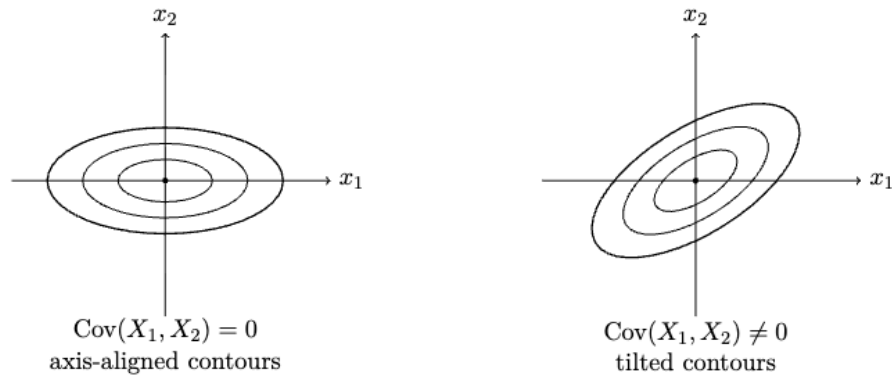


Figure 2: Schematic level sets of two-dimensional Gaussian densities. For jointly Gaussian coordinates, zero covariance implies independence.

11 Markov and Chebyshev inequalities

Definition: Markov's inequality

Let X be a non negative random variable, then.

$$P(X \geq a) \leq \frac{\mathbb{E}[X]}{a}$$

Proof: Markov's inequality

Let X be a nonnegative discrete random variable

$$\begin{aligned} \mathbb{E}[X] &= \sum_x x \mathbb{P}(X=x) \\ \mathbb{E}[X] &= \sum_{x \geq 0} x \cdot P(X=x) \\ &= \sum_{x=0}^a x \cdot P(X=x) + \sum_{x \geq a} x \cdot P(X=x) \\ &= \sum_{x \geq a} x \cdot P(X=x) \geq a \sum_{x \geq a} P(X=x) \\ &= a \mathbb{P}(X \geq a) \\ \frac{\mathbb{E}[X]}{a} &\geq \mathbb{P}(X \geq a) \quad \Leftrightarrow \quad \boxed{P(X \geq a) \leq \frac{\mathbb{E}[X]}{a}} \end{aligned}$$

□

Definition: Chebyshev's inequality

Let X be a non negative random variable, then.

$$P(|X - \mu| \geq k) \leq \frac{1}{k^2}.$$

Proof: Chebyshev's inequality

Let $\mu = \mathbb{E}[X]$ and $\sigma^2 = \text{Var}[X]$. For $k > 0$, then

$$P(|X - \mu| \geq k) \leq \frac{1}{k^2}.$$

Notice that $|X - \mu| \geq k \iff (X - \mu)^2 \geq k^2$

Then by setting $Y = (X - \mu)^2$ we can apply markov's inequality.

$$\begin{aligned} \mathbb{P}(|X - \mu| \geq k) &= \mathbb{P}((X - \mu)^2 \geq k^2) \\ &\leq \frac{\mathbb{E}[(X - \mu)^2]}{k^2 \sigma^2} \quad (\text{applying Markov}) \\ &= \frac{\sigma^2}{k^2 \sigma^2} \\ &= \boxed{\frac{1}{k^2}} \end{aligned}$$

Thus Chebyshev's inequality is simply Markov's inequality applied to the squared deviation from the mean. This essentially mean: The probability that X is more than k standard deviations away from the mean is at most $\frac{1}{k^2}$

12 Exercises

2.7 For any random variables X, Y with finite expectations and constants a, b , show that:

$$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$$

Importantly, X and Y do *not* need to be independent.

Given that

$$\mathbb{E}[aX + bY] = \sum_x \sum_y (ax + by) p_{XY}(x, y)$$

Then

$$\begin{aligned} \sum_x \sum_y (ax + by) p_{XY}(x, y) &= a \sum_x x \sum_y p_{XY}(x, y) + b \sum_y y \sum_x p_{XY}(x, y) \\ &= a \sum_x x \mathbb{P}(X = x) + b \sum_y y \mathbb{P}(Y = y) \\ &= a\mathbb{E}[X] + b\mathbb{E}[Y] \end{aligned}$$

2.8 Suppose there are n students. Each student independently answers a given question correctly with probability p .

1. What is the expected number of students who answer the question correctly?
2. A teaching assistant checks the answers one by one, stopping as soon as a correct answer is found, or after all n answers have been checked. What is the expected number of answers that the teaching assistant checks?
3. Now suppose the instructor randomly chooses between two possible questions:
 - an *easy* question, chosen with probability q , for which each student answers correctly with probability p_E ;
 - a *hard* question, chosen with probability $1 - q$, for which each student answers correctly with probability p_H .

Let X denote the total number of students who answer correctly. Compute $\mathbb{E}[X]$ using conditional expectations and the law of total expectation.

1. $X_i = \begin{cases} 1 & \text{if student } i \text{ answers correct} \\ 0 & \text{otherwise} \end{cases}$

Given that $T = \sum_{i=0}^n x_i$

$$\text{Then } \mathbb{E}[T] = \mathbb{E}\left[\sum_{i=0}^n x_i\right] = \sum_{i=0}^n \mathbb{E}[x_i] = \sum_{i=0}^n p = np$$

2. $X_i = \begin{cases} 1 & \text{if student } i \text{ is checked} \\ 0 & \text{otherwise} \end{cases}$

$$\text{Then } P(x_i = 1) = (1 - p)^{i-1}$$

$$\mathbb{E}[T] = \sum_{i=0}^n \mathbb{E}[x_i]$$

1. $X_i = \begin{cases} 1 & \text{if student } i \text{ is correct} \\ 0 & \text{otherwise} \end{cases}$

$$\text{Then } \mathbb{E}[x_i] = \mathbb{E}[\mathbb{E}[X|Q]] = \mathbb{E}[x_i | E]p_E + \mathbb{E}[x_i | H]p_H = p_E q + p_H(1 - q)$$

$$\mathbb{E}[T] = \sum_{i=0}^n \mathbb{E}[x_i] = n(p_E q + p_H(1 - q))$$

2.9 Let X be uniform on $\{-1, 0, 1\}$ and $Y = X^2$. Verify directly from the joint distribution that X and Y are not independent, but are uncorrelated.

Firstly, independence means that for $p_{XY}(x, y) = p_X(x)p_Y(y)$

Given that $(x, y) = (0, 1)$ that means that

$$\begin{aligned} p_{XY}(0, 1) &= 0 \\ p_X(0)p_Y(1) &= \frac{1}{3} \cdot \frac{2}{3} = \frac{2}{9} \end{aligned}$$

so $p_{XY}(x, y) \neq p_X(x)p_Y(y)$ and X, Y must be dependent.

For uncorrelation, the joint pmf is nonzero only at the three pairs $(-1, 1), (0, 0), (1, 1)$, each with mass $\frac{1}{3}$. So

$$\mathbb{E}[XY] = \sum_x \sum_y xy p_{XY}(x, y) = (-1)(1)\frac{1}{3} + (0)(0)\frac{1}{3} + (1)(1)\frac{1}{3} = 0$$

$$\mathbb{E}[X] = \sum_x x p_X(x) = (-1)\frac{1}{3} + 0 \cdot \frac{1}{3} + 1 \cdot \frac{1}{3} = 0$$

Therefore

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = 0 - 0 \cdot \mathbb{E}[Y] = 0$$

Thus X, Y are uncorrelated (despite being dependent).

2.10 A disease affects 1% of a population. A test is positive with probability 0.99 when the disease is present and with probability 0.05 when it is absent. Compute the probability that a person has the disease given a positive test.

Given that

- $P(D) = \frac{1}{100}$
- $P(+ | D) = \frac{99}{100}$
- $P(+ | \neg D) = \frac{5}{100}$

Then

$$P(D | +) = \frac{P(+ | D)P(D)}{P(+)} = \frac{P(+ | D)P(D)}{P(+ | D)P(D) + P(+ | \neg D)P(\neg D) +}$$

2.11 Let $X \sim \mathcal{N}(0, 1)$ and define $Y = 2X + 3$. What are $\mathbb{E}[Y]$ and $\text{Var}(Y)$? What is the distribution of Y ?

Given that

- $X \sim \mathcal{N}(0, 1)$
- $\mathbb{E}[X] = 0, \text{Var}[X] = 1$

Then $\mathbb{E}[Y] = 2 \cdot \mathbb{E}[X] + \mathbb{E}[3] = 0 + 3,$

And $\text{Var}[Y] = \text{Var}[2X] + \text{Var}[3] = 2^2 \text{Var}[X] + 0 = 4$